



2026 BASIC ABSTRACT ALGEBRA STUDY

LECTURE 2 HOMEWORKS

SOLUTION - Q5

Version

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Note 1 (Notations)

The following notations are used throughout the questions below.

- $\mathbb{Z}$: The set of all integers.
- $\mathbb{Z}^+$: The set of all positive integers; i.e., $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$.
- $\mathbb{Z}_n$: The set of non-negative integers from 0 to $n - 1$ for some positive integer n . For instance, $\mathbb{Z}_4 = \{0, 1, 2, 3\}$.
- $\mathbb{R}$: The set of all real numbers.
- $\min(x, y)$: The minimum of x and y ; i.e., the smaller of the two.
- $\mathcal{P}(X)$: The power set of the set X .
- μ_n : The set of all n -th roots of unity.

Q1.

EASY

Let $f, g, h : \mathbb{Z} \rightarrow \mathbb{Z}$ be given by

$$f(x) = x + 3, \quad g(x) = 2x, \quad h(x) = x - 1.$$

What is the value of $(h \circ g \circ f)(1)$?

- ☐ 1. 3
 ☐ 2. 5
 ☐ 3. 7
 ☐ 4. 8
 ☐ 5. 9

Q2.

EASY

Let S_5 be the set of all bijections from $\mathbb{Z}_5$ to $\mathbb{Z}_5$.

What is the value of $|S_5|$?

- ☐ 1. 5
 ☐ 2. 25
 ☐ 3. 120
 ☐ 4. 125
 ☐ 5. 625

Definition 2 (Injection)

A function $f : X \rightarrow Y$ is said to be *injective*, or called an *injection*, if for every $x, y \in X$ with $x \neq y$, we have $f(x) \neq f(y)$.

Q3.

EASY

Classify each of the following functions as injective or NOT injective. Which one differs from the rest?

- ☐ 1. $f_1 : \mathbb{R} \rightarrow \mathbb{R}, f_1(x) = \min(x, 2x)$
☐ 2. $f_2 : \mathbb{R} \rightarrow \mathbb{R}, f_2(x) = 0 \cdot x$
- ☐ 3. $f_3 : \mathbb{Z} \rightarrow \mathbb{Z}, f_3(x) = \gcd(x, 10)$
☐ 4. $f_4 : \mathbb{R} \rightarrow \mathbb{R}, f_4(x) = x^2 + 5x + 6$
- ☐ 5. $f_5 : \mathbb{Z} \rightarrow \mathbb{Z}, f_5(x) = |x|!$

Definition 3 (Equality of functions)

Two functions with the same domain and same codomain are said to be *equal* if and only if they map every element of the domain to the same element of the codomain.

In other words, if $f, g : X \rightarrow Y$ are functions with the same domain and codomain, then $f = g$ if and only if $f(x) = g(x)$ for every $x \in X$.

Q4.**EASY**

Given that every function below has $\mathbb{Z}^+$ as both its domain and codomain, which one differs from the rest?

- ☐ 1. $f_1(x) = \gcd(x, x^2) + \gcd(x, x^3)$
☐ 2. $f_2(x) = \text{lcm}(2x, 3x) - \text{lcm}(x, 3x)$
- ☐ 3. $f_3(x) = {}_{x+1}C_x + x - 1$
☐ 4. $f_4(x) = \gcd(2x, 6x)$
- ☐ 5. $f_5(x) = |3x| - |x|$

Definition 4 (Idempotent function)

A function f from a set X to itself is called *idempotent* when $f \circ f = f$.

Q5.**EASY**

Which of the following functions is NOT idempotent?

- ☐ 1. $f_1 : \mathbb{Z} \rightarrow \mathbb{Z}, f_1(x) = x \bmod 5$
☐ 2. $f_2 : \mathbb{R} \rightarrow \mathbb{R}, f_2(x) = \min(\sqrt{2}, x)$
- ☒ 3. $f_3 : \mu_{64} \rightarrow \mu_{64}, f_3(x) = x^8$
☐ 4. $f_4 : \mathbb{R} \rightarrow \mathbb{R}, f_4(x) = |2x| - |x|$
- ☐ 5. $f_5 : \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{P}(\mathbb{R}), f_5(x) = x \cup \mathbb{Z}$

- ☐ The value $f_1(x) = x \bmod 5$ lies in $\{0, 1, 2, 3, 4\}$, and each of these is its own remainder modulo 5. So applying f_1 again changes nothing: $f_1(f_1(x)) = f_1(x)$. Thus, f_1 is idempotent.
- ☐ Since $f_2(x) = \min(\sqrt{2}, x) \leq \sqrt{2}$, we have $f_2(f_2(x)) = \min(\sqrt{2}, f_2(x)) = f_2(x)$. Thus, f_2 is idempotent.
- ☒ Take a root of unity $\zeta = e^{2\pi i/64}$. Then $f_3(\zeta) = \zeta^8 = e^{2\pi i/8}$, while $f_3(f_3(\zeta)) = \zeta^{64} = 1$. These differ, so f_3 is not idempotent.
- ☐ Since $|2x| = 2|x|$, we have $f_4(x) = 2|x| - |x| = |x|$. As $|x| \geq 0$, applying f_4 again gives $f_4(f_4(x)) = ||x|| = |x|$. Thus, f_4 is idempotent.
- ☐ For a subset x of $\mathbb{R}$, $f_5(x) = x \cup \mathbb{Z}$. Then $f_5(f_5(x)) = (x \cup \mathbb{Z}) \cup \mathbb{Z} = x \cup \mathbb{Z} = f_5(x)$, since $\mathbb{Z} \cup \mathbb{Z} = \mathbb{Z}$. Thus, f_5 is idempotent.

Therefore, the third choice is the correct answer.

End.

You may now submit your work.

Optional evaluation sheet

Unlike PCSA, submitting the sheet is **optional** for homeworks.

For each question so far, you can evaluate it based on the following criteria:

- If you failed to understand the question, mark $\times$ in the Evaluation column.
- If you understood the question, but didn't know how to solve it, mark ? in the Evaluation column.
- If you understood the question and are sure you have solved it, mark $\circ$ in the Evaluation column.

No.	Evaluation	No.	Evaluation
Q1		Q4	
Q2		Q5	
Q3			

There are also a few additional questions; feel free to answer them.

- (a) Which was the hardest for you, among those you have solved? _____
- (b) Which seemed to be the hardest for you, among those you haven't solved? _____
- (c) Which was your favorite question, if you had to choose one? _____